Data Driven Computer Vision Oriented Image Signal Processing Pipeline Configuration

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Abstract

The image signal processing (ISP) pipeline in camera systems is traditionally designed for human vision. A human vision-oriented ISP pipeline is optimized independently of the backend vision algorithms for computer vision systems, which may lead to a suboptimal solution for the entire system. To find the optimal configuration of the ISP pipeline for vision tasks, in this paper, we propose a hybrid supervised learning and reinforcement learning-based framework to co-design these two parts in a unified framework. To achieve this, an ISP policy network, ISPNet, is designed to output the vision performance-oriented ISP configuration with a given input RAW image for a specific computer vision task. In particular, the ISPNet is trained together with a vision task model, where the output policy aims to maximize a vision performance-related reward while bypassing as many ISP steps as possible (for overall computational complexity reduction). Extensive experiments based on classification and object detection are conducted. The results show that the entire ISP pipeline, or most ISP stages, can be eliminated if the following consumer is computer vision tasks instead of human eyes. Models trained on RAW images are robust enough for tasks such as classification and object detection under most normal image shooting conditions. It is also shown that

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the demosaicing step is necessary for tasks with dedicated pre-processing and/or data augmentation since it can resist interpolation artifacts and improve the generalization ability of models. While under extreme shooting conditions, the proposed dynamic configuration framework can outperform a fixed policy in model performance.
Keywords: Image signal processing (ISP), computer vision, co-design, associative search

1. Introduction

Computer vision studies how to enable computers to understand imagery inputs such as photos and videos. A typical computer vision system consists of a frontend imaging system, i.e., a camera, and a backend vision system running specific vision algorithms. The majority of modern computer vision algorithms take finely rendered color images as input, most of which are produced by cameras equipped with one image sensor [1]. For these single-sensor cameras, the scenes are captured by an image sensor covered with a color filter array (CFA) to maintain the color information. The output RAW images are then processed by a series of photography-oriented image processing algorithms, known as the image signal processing (ISP) pipeline, to produce finely rendered color images for human eye consumption. That is, the RAW images have already included all the scene-related information, and the purpose of the ISP pipeline is to improve the presentation of these captured contents to make them more suitable for the human vision system (HVS). For example, gamma compression [2] is designed to satisfy the human perception of brightness (Stevens' Power Law [3]), and white balance is used for restoring the white objects to white under different light sources (color constancy [4]). Therefore, as we can see, the introduction of the ISP pipeline is to adapt the captured information to the characteristics of HVS instead of computer vision tasks. Though pleasing images can be generated, these visual-enhance ISP operations may be harmful to computer vision tasks. For example, many computer vision tasks assume cameras are accurate light measurement devices, implying that the pixel values are linearly related to the radiance of real scenes[5]. However, with the development of photography technology and the continuous increment of people's

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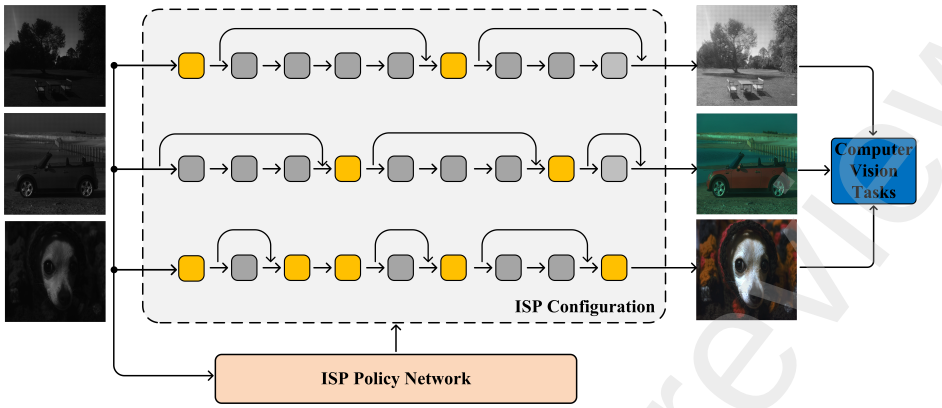

Figure 1: Illustration of ISP pipeline configuration for various computer vision tasks.

aesthetic requirements, more and more sophisticated processing steps with nonlinear operations are introduced to the ISP pipelines, leading to an even more severe nonlinear correlation between pixels and scenes. Moreover, apart from the suitability for computer vision tasks, the computational complexity of sophisticated ISP pipelines may also affect the overall power, delay and price of vision systems.

For most of the existing computer vision systems, the frontend camera system and backend vision system are designed and optimized individually to achieve their own best performance. Co-designing camera systems with vision algorithms to improve the end-to-end performance and efficiency of the entire computer vision system is an interesting and challenging problem[6]. Conventional ISP pipelines are highly modular, efficient and hardware friendly, where each algorithm in the pipeline is designed to handle different practical imaging situations and has corresponding efficient hardware implementation. To improve the vision performance and computational efficiency of existing vision systems, this paper explores the potential of dynamically allocating processing steps in conventional ISP pipelines, for given computer vision applications. In particular, we propose to learn the vision application-specific ISP configuration as illustrated in Fig. 1, where an ISP policy network is trained together with the vision application network to dynamically determine the necessary but minimal processing steps according to the input RAW images.

To achieve this, a decision module needs to be constructed and co-optimized with

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Table 1: A 10-stage ISP pipeline used in this work

ISP Steps	Functionality
Linearization	To transform the RAW data into linear space.
Dead Point Correction	To correct the dark, flare, and other abnormal pixels in RAW images.
Anti-aliasing Filtering	To reduce aliasing artifacts for further processing.
Bayer Denoise	To suppress noises in the RAW domain.
Demosaicing	To interpolate single-channel Bayer pattern images into three-channel color images.
White Balance	To remove unrealistic color casts such that white objects are rendered white.
Gamma Correction	To compress the dynamic range of images while preserving the visual effect.
Sharpen	To enhance the edges for clarity improvement.
Histogram Equalization	To adjust image contrast by equalizing the distribution of image pixel values.
RGB Denoise	To suppress noises introduced in preceding steps.

the vision tasks together, aiming to correctly bypass the unnecessary processing steps. However, this inherently discrete decision (bypassing a certain step or not), as well as the image processing algorithms in the ISP pipeline, are not differentiable, meaning that the gradient-based optimization methods can not be directly applied. Though there are re-parameterization methods for non-differentiable operations, such as the Gumbel Softmax trick [7] which allows the gradients to backpropagate through a discrete sampling process, or the approach of using differentiable soft-max decision in the training process and then revert to hard decisions during inference[8], the approximation introduced by relaxation in these methods often stuck the model into suboptimal solutions [9].

Inspired by the works on the dynamic route and layer skipping of convolutional networks [9, 10], in this paper, we propose a hybrid learning algorithm, ISPNet, which combines supervised learning with deep reinforcement learning to obtain a vision performance-oriented ISP configuration policy for a given input RAW image and a specific computer vision task. In particular, the output of ISPNet is a binary decision vector (whose elements are either 0 or 1) with a length equal to the number of processing steps in the ISP pipeline, indicating the corresponding steps should be bypassed (0) or not (1). The ISPNet is trained together with a vision task model, where the target of the output policy is to maximize a vision performance-related reward while bypassing as many ISP

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steps as possible, so as to reduce the computational complexity of the vision system.

To validate the proposed method, extensive experiments based on two popular vision tasks, classification and object detection, are conducted in this work. The results show that features extracted from RAW Bayer pattern images are robust enough for follow-up computer vision tasks, and the ISP pipeline can be bypassed completely in most cases. Our trained ResNet50 model achieves 78.21% top-1 accuracy on the transformed RAW ImageNet dataset and comparable object detection performance. Moreover, for tasks that have dedicated pre-processing or data augmentation technologies, the demosaicing stage can eliminate the artifacts caused by interpolation in these operations, and boost the generalization ability of the models. For images shot under extreme conditions such as low-light, our proposed dynamic framework outperforms a fixed policy in model performance. The main contributes of this work are:

- A hybrid reinforcement and supervised learning-based dynamic configuration framework to obtain the optimal ISP configuration policy for a specific computer vision task.
- Detailed guidelines for ISP configuration under different shooting conditions.
- An end-to-end tool for RGB to RAW image translation to generate training data for ISP-computer vision co-design research.

2. Related Work

2.1. Traditional ISP pipeline

The traditional ISP pipeline is a highly modular step-by-step image processing chain. This kind of pipeline is commonly used in mono-sensor color camera techniques, which is the mainstream solution for digital cameras and smartphones nowadays[1]. Generally, an ISP pipeline covers functions such as camera correction, interpolation, color space conversion and enhancement, though the specific implementations of these functions may vary for different camera manufacturers. Recently, there are also many works on designing deep learning-based end-to-end ISP pipelines for high-quality images[11], especially for demosaicing and denoise [12].

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Since traditional ISP pipelines with efficient dedicated hardware implementations are widely used in modern digital cameras, we focus our research on conventional ISP pipelines to make the proposed approach compatible with existing systems[13]. In this paper, we use a 10-stage ISP pipeline shown in Table 1 in our study. Note that the steps included are commonly used image processing operations in ISP, other steps can be configured into the proposed framework in the same manner to study their effect on deep learning-based computer vision tasks.

2.2. ISP pipeline for Computer Vision

The optimal configuration of computer vision-oriented ISP pipeline remains an open problem, attracting a lot of attention in recent years. To bridge the gap between the ISP design and computer vision tasks, several pioneer works try to locate and bypass the redundancies in conventional ISP pipelines when they are used in vision applications. By analyzing the way how hand-craft features such as histogram of oriented gradients (HOG), scale-invariant feature transform (SIFT) and deformable parts models (DPM)[14] are extracted, some works try to bypass the entire ISP pipeline and extract features from RAW images directly or cross-domain [15, 16]. For the deep learning-based vision algorithms, Buckler et. al. conduct extensive experiments on eight commonly used models and outline a minimal ISP pipeline in [17]. Although this kind of brute-force experimental method can arrive at a conclusion for a specific vision algorithm, it cannot be directly extended to other vision algorithms. Moreover, it takes a lot of time and resources to experiment with a new algorithm, and the experimental efforts increase exponentially with the number of ISP steps in the pipeline.

[18] proposes a framework called ρ-Vision, which enables direct execution of computer vision tasks in the RAW domain, thereby bypassing the traditional image signal processor (ISP) pipeline. In addition, the authors design an inverse ISP model to generate simulated RAW data (simRAW) from RGB images, addressing the scarcity of real RAW datasets. However, the study does not thoroughly investigate the potential benefits that ISP might offer for visual tasks under extreme conditions, such as low light or high noise scenarios. In [19], Yu et. al. propose a reconfigurable ISP pipeline whose architecture and parameters can be tailored for individual data and tasks automatically.

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In particular, they train one proxy network for each algorithm in the ISP pipeline, and then replace these proxy networks with the corresponding ISP algorithms throughout the training process. By introducing the proxy networks, the non-differentiable problem of ISP algorithms is solved, and these proxy networks can be trained with computer vision tasks while improving the image quality. However, after several rounds of parameter updating, the trainable proxy networks may lose functional coherence with the associated ISP algorithms, and it is challenging to modularize for hardware implementation. In addition, to ensure the visualization quality of images, unnecessary steps may be retained after reconfiguration. As a result, in this paper, we preferred to maintain the original ISP methods in the training process in order to be compatible with the practical application scenarios.

2.3. RAW Image Datasets

Deep learning-based algorithms usually require a large number of training images. However, since most of the modern computer vision algorithms consume finely rendered color images, there are few publicly available RAW image datasets for computer vision tasks. There are some popular RAW datasets for image processing studies such as image restoration, enhancement[20] and denoise[21]. For example, the Kodak true color image dataset [22], the Laurent Condat's (LC) image base[23] and the McMaster dataset[24], are popular in end-to-end ISP design or evaluation of algorithms in traditional ISP pipelines, especially demosaicing. However, all these HVS-oriented tasks are aimed at improving the visual quality of images instead of vision tasks' performance. There are also some small-scale RAW datasets for tasks such as object detection [14, 25], but the volume of these datasets may be too small for deep learning-based algorithms. Although commercial photography devices such as mobile phones and digital single-lens reflex (DSLR) cameras can be configured to save RAW images, collecting and annotating a large-scale dataset would require a tremendous amount of effort.

As an alternative, there are studies that try to generate RAW images from the corresponding color images by reverting the ISP pipeline [26, 17] or using end-to-end learning-based methods [27]. In this paper, we follow the suggestions in [27] to syn-

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thesize the RAW Bayer pattern images from existing color image datasets, details are discussed in Sec. 3.1.

2.4. Dynamic Neural Networks

Dynamic networks are designed to adapt their structures or parameters to the input during inference, and therefore have advantages in efficiency, representation ability, adaptiveness, compatibility, generality and interpretability over static models [28, 29]. Among these works, layer-skipping approaches aim to set up a dynamic path in network inference by bypassing unnecessary intermediary layers [9, 10]. The authors of [9] propose a hybrid learning method to learn skip policy. They introduce a gating module for each layer in ResNet, which outputs a 0 or 1 decision respectively to decide whether a specific layer should be bypassed or applied. The vector composed of these $\{0, 1\}$ decisions is the skipping policy of the corresponding RAW input. Different from standard reinforcement learning, [10] trains a policy network to obtain all the skip decisions for a specific input at once, i.e., the policy network output a binary vector directly. All these methods optimize the task performance and skipping policy together, where the policy network outputs the hard binary decision.

Inspired by the aforementioned works, if the steps or transformations in an ISP pipeline can be embedded into the vision algorithm frameworks as a fixed layer, then a similar policy network can be trained to optimize a vision task-specific ISP configuration.

3. Methodology

3.1. Dataset Preparation

As mentioned in Sec. 2.3, in order to prepare adequate training data for deep learning tasks, we propose to convert color image datasets to their corresponding RAW version using a Generative Adversarial Network (GAN). As shown in [27], under a set of invertible and differentiable transformations $T=\{T_1, T_2,..., T_k\}$, the virtual training criterion of GAN can be expressed as

$$V(G, D_T^*) = -log(4) + 2 \cdot JS(p_{\text{data}} \parallel p_{\text{g}}), \tag{1}$$

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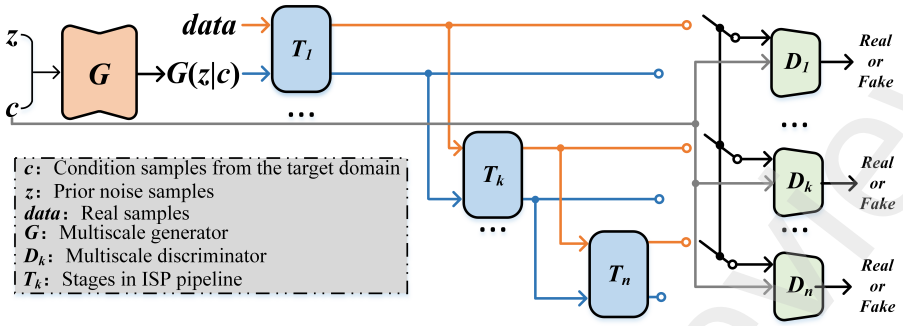

Figure 2: Overall architecture of the color-to-RAW GAN.

(a) (b)

Figure 3: (a) The original images from the ImageNet dataset and (b) the corresponding synthesized RAW images after applying the ISP pipeline in Table 1.

Equation (1) indicates that given an optimal discriminator D_T^* trained with transformed samples, the objective is to minimize the Jensen-Shannon (JS) divergence between the real samples and generated raw samples (instead of the generated samples with transformation[27]). As a result, some un-formulatable constraints can be added in GAN training to obtain desired data. For example, adding the demosaicing step ensures that the output images follow the RAW Bayer pattern image distribution and could be demosaiced.

For the RGB-to-RAW conversion module, we adopt the overall GAN framework proposed in [27], including the multi-scale generator and discriminator design. To

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further enhance the inversion process, we additionally introduce a differentiable white balance transformation into the mapping set T as suggested in [27]. The complete architecture is illustrated in Fig.2. To prepare the training data, we combine the datasets from [30, 14, 31, 32] to produce 14033 RAW-RGB image pairs. The GAN model for the final RGB to RAW transformation is trained for more than 70k steps. Fig. 3 shows a number of examples that are first converted from the original ImageNet images (Fig. 3(a)) to their corresponding RAW Bayer domain, and then processed by the 10-stage ISP pipeline in Table 1 to convert back to color (Fig. 3(b)). As we can see, the original and synthesized images are visually identical in terms of content and image quality, the difference in color is caused by different ISP pipelines. The original images are processed by the ISP pipeline in the camera, while the synthesized images are processed by the pipeline in Table 1.

To better simulate real-world degraded conditions (e.g., low-light, overexposure, noise), we introduce synthetic degradations to the RAW images. For example, gamma distortion is applied to model nonlinear exposure shifts, which can be formulated as (3)

$$I_{degraded} = I^{\gamma}, \tag{2}$$

where $\gamma \in [0.5, 4]$, lower values ($\gamma < 1$) simulate overexposure by enhancing shadows and suppressing highlights, while higher values($\gamma > 1$) emulate underexposed or dim environments. Additive Gaussian noise is used to simulate sensor noise, represented as

$$I_{degraded} = I + N(0, \sigma^2), \tag{3}$$

with $\sigma \in [0.01, 0.05]$.

3.2. ISP Configuration Framework

As previously discussed, our objective is to find the optimal input-specific ISP configuration for a certain computer vision task. In this work, an extra module is designed to generate the 'policy' for each input RAW image, and this module is optimized together with computer vision tasks. Fig. 4 illustrates the overall structure of the proposed ISP configuration framework. As we can see, for a given N-step ISP pipeline, the RAW input is first fed into the policy network, i.e., ISPNet, to obtain an N-dimension

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decision vector. Then the RAW image along with the decision vector is input to the ISP pipeline to obtain a processed image under the specific ISP configuration generated by ISPNet. The decision vector, consisting of 0 and 1 elements, indicates whether the stages in the ISP pipeline should be bypassed or not. After that, the output image is passed to the vision task model, and the framework is updated according to the performance of the vision task and ISP configuration results.

Since the decision, generated by ISPNet, for each ISP stage is essentially discrete (either 0 or 1), the continuous gradient-based method will not work. According to the characteristics of ISP pipelines, we have the following considerations:

- The complexity of the algorithms in the ISP pipeline is quite modest. The majority of the processes in the ISP pipeline, unlike feature extraction, are global transformations, meaning that the content of images will not change and the difference is primarily visual. Therefore, designing a gate module for each step appears to be superfluous. Furthermore, when the number of ISP steps increases, the number of gate modules increases accordingly, which is inconvenient for model training and generalization.

- Each step in the ISP pipeline is independent. The traditional ISP pipeline is highly modular. In most cases, each module performs its own function and cannot be replaced by others. As a result, the decision to skip one step should not be influenced by the preceding step. Instead, it should be determined by the original inputs (the RAW images).

- The structure of ISP pipelines is relatively fixed. Although ISP pipelines allow for flexibility in algorithm sequencing, some steps must be completed in a specific order, and algorithms must be applied in a proper domain (Bayer domain or RGB domain). In addition, algorithms that have already been applied will not be repeated. Therefore, in this paper, we only discuss the fixed pipeline as shown in Table 1. That is, whether one step is bypassed or not, the next state in policy learning is the following step, i.e., the order of the ISP steps will not change.

According to the above analysis, it can be found that although the decision (action)

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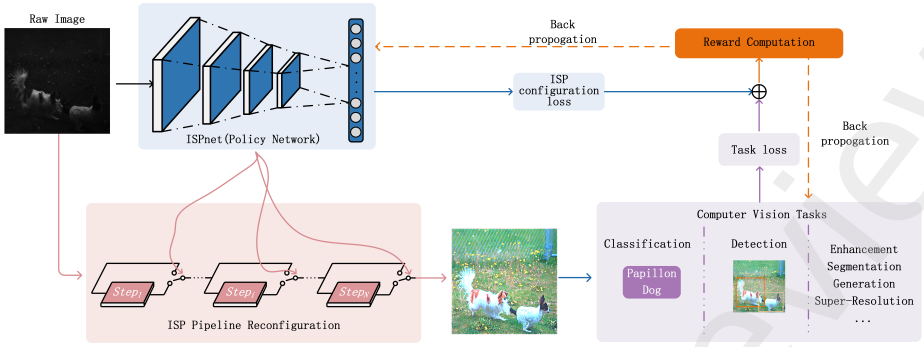

Figure 4: The overall architecture of the ISP configuration framework.

is associated with the current step (situation), the action taken by this step will not affect the next step. Taking the ISP pipeline in Table 1 as an example, for any input, the policy network will give a corresponding action to demosaicing, regardless of the decision value, the next step must be white balance.

Therefore, this process can be regarded as an associative search problem, because it involves both searching for the best actions and associating these actions with the situation in which they are best, and these actions are not allowed to affect the next situation [33]. Since all the actions are binary and independent of one another (independent and identically distributed), the output policy of the ISP configuration policy network, i.e., ISPNet, behaves as an N-dimension Bernoulli distribution. As a result, the output of ISPNet is an N-dimension vector that can be expressed as

$$f_{\text{isp}}(x, \theta_p) = [g_1, g_2, ..., g_N], \tag{4}$$

where $f_{\text{isp}}(\cdot)$ is the ISPNet parameterized by the weights θ_p. The valuable g_i, processed by the sigmoid function, i.e., $g_i \in [0, 1]$, represents the probability of $a_i = 1$. In our design, the proposed policy network consists of four residual blocks (Fig. 5) , whose parameter quantity is far less than the computer vision tasks.

Then the output distribution of ISPNet can be expressed as

$$\begin{cases} P(a_i = 1 \mid x) = g_i \\ P(a_i = 0 \mid x) = 1 - g_i, \end{cases} \tag{5}$$

Thus, the output policy of ISPNet is

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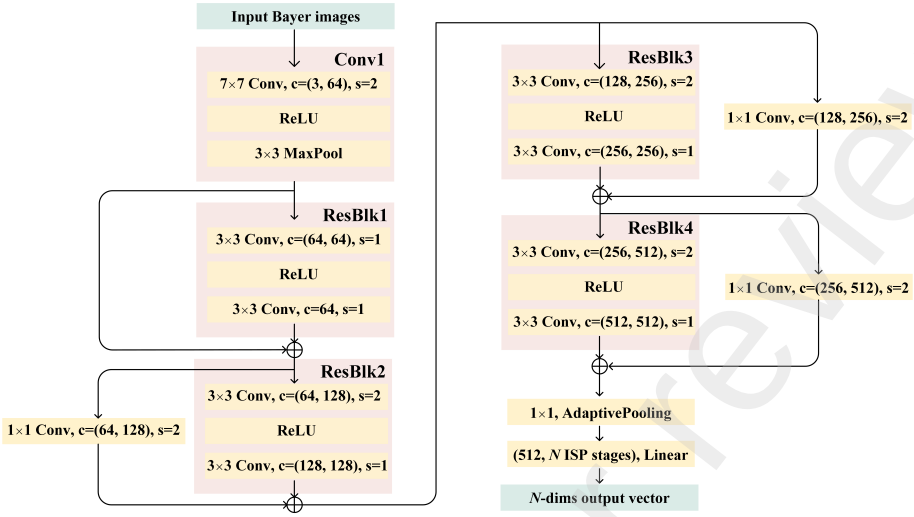

Figure 5: The overall architecture of the policy network.

$$\pi\left(A \mid x, \theta_p\right) = \prod_{i=1}^{N} g_i^{a_i} \left(1 - g_i\right)^{1-a_i}. \tag{6}$$

The decision value $a_i \in \{0, 1\}$ is sampled based on g_i, where $a_i = 1$ indicates the i-th step in ISP pipeline should be executed, while $a_i = 0$ denotes the step should be bypassed. Let f_{s_i} represents the i-th step in ISP pipeline, the output of f_{s_i} can be expressed as

$$\mu_{i+1} = a_i \cdot f_{s_i}(\mu_i) + (1 - a_i) \cdot \mu_i. \tag{7}$$

Here μ_i is the input to the i-th ISP step, $\mu_1 = x$ is the original RAW input.

To encourage the minimal usage of ISP steps, the actions taken by the policy network are rewarded by the following function

$$R(A) = 1 - \frac{\sum_{i=1}^{N} a_i w_i}{N}, \tag{8}$$

which gives the proportion of the steps in the ISP pipeline being skipped. Here, w_i is the cost of executing the $i-$th step, which ranges from $[0, 1]$ and can be determined by the measurements such as the complexities of the corresponding algorithms.

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3.3. ISPNet Training with Vision Tasks

As mentioned before, our goal is to configure the ISP pipeline with the specific computer vision tasks, hence, these two networks should be optimized together. To achieve this, the reward should be associated with the performance of the computer vision tasks, presented as

$$R(A) = \begin{cases} \lambda_r \left(1 - \frac{\sum_{i=1}^{N} a_i w_i}{N} \right), & \text{meet the conditions} \\ \lambda_p. & \text{otherwise} \end{cases} \tag{9}$$

Here, λ_r and λ_p are the reward factors. When the computer vision tasks fulfill the performance requirements, e.g., the classification is correct, or the detection achieves the required accuracy, the more steps being skipped, the higher the reward will be. Otherwise, a λ_p penalty will be given. The goal of the policy network is to maximize the expected reward, expressed as

$$J(\theta) = v_{\pi(\theta_p)}(x) \tag{10}$$

Therefore, the overall object function is given by

$$min\mathcal{J}(\theta) = min\mathbb{E}\left[\mathcal{L}_{\theta_t}(f(x), y) - v_{\pi(\theta_p)}(x) \right]. \tag{11}$$

Here, $\mathcal{L}_{\theta_t}(f(x), y)$ is the loss function of the computer vision task parameterized by θ_t. The gradients can then be derived similarly as

$$\begin{aligned} \nabla_\theta \mathcal{J} &= \mathbb{E}\left\{ \nabla_{\theta_t} \mathcal{L} - R(A) \nabla_{\theta_p} ln\left[\pi(A \mid x, \theta_p) \right] \right\} \\ &= \mathbb{E}\left\{ \nabla_{\theta_t} \mathcal{L} - R(A) \nabla_{\theta_p} \sum_{i=1}^{N} \left[a_i ln g_i + (1 - a_i) ln(1 - g_i) \right] \right\}. \end{aligned} \tag{12}$$

We approximate the expected gradient using the classic REINFORCE algorithm, which employs Monte Carlo estimation of policy gradients[34]. Although these gradient estimates are unbiased, they typically exhibit high variance, which can lead to slow convergence and instability during training[33]. To improve training stability, we introduce a baseline function and rewrite the objective as follows:

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$$\begin{aligned}\nabla_\theta \mathcal{J} &= \mathbb{E}\left\{\nabla_{\theta_l}\mathcal{L} - \mathcal{A}\nabla_{\theta_p} ln\left[\pi(A \mid x, \theta_p)\right]\right\} \\ &= \mathbb{E}\left\{\nabla_{\theta_l}\mathcal{L} - \mathcal{A}\nabla_{\theta_p}\sum_{i=1}^{N}[a_i ln g_i + (1-a_i) ln(1-g_i)]\right\}.\end{aligned} \tag{13}$$

Here, $\mathcal{A} = R(A) - b(s)$, where $b(s)$ is the running mean of $R(A)$ and update by $b_t(s) = \beta R(A) + (1-\beta)b_{t-1}(s)$.

Moreover, to encourage further exploration [35], we add a self-entropy term to (11), which can be expressed as

$$\mathcal{L}_{\theta_p} = \alpha \sum_{i=1}^{N} g_i ln(g_i), \tag{14}$$

where α is the weight parameter for the self-entropy.

4. Datasets & Experimental Setup

4.1. Datasets

We evaluate our method on two datasets, the synthesized ImageNet-RAW dataset (described in Sec. 3.1) for classification and the PascalRAW dataset for object detection [25]. In the classification task, the converted images are all 448×448 single-channel RAW *RGGB* Bayer pattern images, while in the object detection task, the PascalRAW dataset contains 4259 RAW *RGGB* images with labeled person, bicycle and car. The original RAW images in the PascalRAW dataset are all 4012×6034 images. Due to the limitation of GPU memory, these images are first fed into the ISPNet to obtain the ISP configuration policy, and then rescaled to the desired size after the corresponding ISP processing to feed into the object detection network.

4.2. Experimental Setup

There are two kinds of computer vision tasks evaluated in this work, image classification and object detection, with the ISPNet trained with the appropriate backbone. For the classification task, we use ResNet50 as the backbone. The model is evaluated on the ImageNet-RAW validation set of 50000 images and learned ISP pipeline configuration policies are then reported. The reward factors in (9) are given by Algorithm 1 described as follows.

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For the object detection task, the Faster-RCNN and Yolo-v3 frameworks are chosen as the detection backbone. These models are tested on the PascalRAW dataset and the COCO mean average precision, AP (the Intersection-over-Union (IoU) from 0.5:0.95), AP_{50} (IoU=0.5) and AP_{75} (IoU=0.75) are reported. The reward is determined by the detection performance p on each image, where p is defined as the ratio of true positives to predicted positives. Since p is a continuous value, the reward is defined as the difference between p and a predetermined threshold. In order to define a reasonable threshold and reduce variance, a model trained on color images is introduced as a reference. Specifically, there is a color counterpart for each RAW Bayer pattern image that is fed into the reference model to get detection precision as a threshold. Algorithm 2 presents the complete reward computation for object decision tasks.

Note that all the weights w_i in (9) are set to 1, the entropy loss weight α in (14) is set to 0.1 in our experiments.

Algorithm 1 Reward factor computation for classification tasks.

Require: Classfication outputs t and ground truth t^*
Ensure: The reward factors in (9)
1: **if** $t == t^*$ **then**
2: return λ;
3: **else**
4: return -λ;
5: **end if**

Algorithm 2 Reward factor computation for object detection tasks.

Require: Model output t, reference model output t' and ground truth t^*.
Ensure: The reward factors in (9).
1: Computing precision p_t and $p_{t'}$
2: **if** $p_t == p_{t'} == 1$ **then**
3: return λ;
4: **else**
5: return $\lambda(p_t - p_{t'})$.
6: **end if**

5. Experimental Results

This section presents the experimental results to demonstrate the effectiveness of the proposed method.

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Table 2: Configuration results of classification tasks

Training Strategy	Configuration	Top-1 acc.	Top-5 acc.	Data Augmentation
Without ISPNet	Keep all	77.81%	94.30%	✓
Scheme 1	Keep white balance and Demosaicing	78.11%	94.33%	✓
	Keep white balance and Demosaicing	77.98%	94.39%	✗
Scheme 2	keep Demosaicing	78.17%	94.46%	✓
	Bypass all	78.21%	94.40%	✗

Data augmentation including random resize, clip, flipping, etc.

5.1. Configuration Results

Configuration Results on Classification. In our training process, the following two training strategies are applied to train the network for classification:

- Scheme 1. Freeze the computer vision task network with pre-trained weights first, and train the ISPNet for 80 epochs. Then unfreeze the task network and jointly fine-tune the entire network including the vision task model and ISPNet for another 20 epochs.
- Scheme 2. Train the entire network including the vision task model and ISPNet from scratch for 100 epochs.

Table 2 presents the configuration results under two different training schemes, note that all configurations were made dynamically. Since there exists random resizing operation in data augmentation, for a fair comparison, the final input images are all set to 448×448. It can be observed from Table 2 that when the model is trained using Scheme 2 without any data augmentation technology, the ISPNet generates the configuration to bypass all ISP stages and the model achieves the best top-1 accuracy among all the configurations, indicating that the features extracted from RAW Bayer pattern images are robust enough for classification. If data augmentation techniques are applied, however, the ISPNet will keep the demosaicing stage. This is because some data augmentation operations, in particular, interpolation-based operations like resize, rotate, etc., will not only destroy the Bayer pattern, but also induce numerous strange artifacts on Bayer pattern images, which seriously affect the model performance.

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Fig. 6 gives an example of resizing a Bayer pattern image to several commonly used sizes in computer vision tasks. Note that the size of the original images is 448×448. It can be found from the top row of Fig. 6 that when the Bayer pattern images are resized to other different sizes, there are different artifacts appear in the image background, which sometimes 'disappear' under special sizes and smooth the images, for example in the case of 224×224 (0.5×scale). However, the difference in the image background after resizing is minor for the demosaiced images (the row below), primarily near the edges (the flowerpot). This is because the interpolation is applied on neighborhood pixels, pixels in demosaiced images vary slowly in the flatten areas, but pixels in RAW Bayer images change dramatically due to the particular alternating pattern. One solution is to resize the images based on *RGGB* super-pixels, meaning that applies interpolation to the pixels from the same channel. However, under a large resize factor, this method will also cause serious aliasing artifacts. Therefore, if the preprocessing and data augmentation include interpolation-like operations, applying the demosaicing stage is an effective way to avoid these artifacts.

For the models trained under Scheme 1, whether they use the data augmentation operations or not, the ISPNet converges to a unified ISP pipeline configuration, which keeps the demosaicing and white balance stage. In particular, in the training process, the ISPNet almost converges in the first epoch, and the model performance continues to improve during the process of jointly fine-tuning the model, but the configuration results remain unchanged. The configuration results indicate that for this ISP pipeline, after applying several global transformations, such as white balance, demosaicing and tone mapping, the distribution of image pixel values approaches the images applied entire ISP to computer vision models. Fig. 7 presents the comparison between the original color image from ImageNet(Fig. 7(a)), the converted image processed by the ISP pipeline in Table 1 (Fig. 7(b)) and the converted image with only demosaicing and white balance (Fig. 7(c)). It can be found from Fig. 7(b) and 7(c) that the difference between these two images is mainly in brightness, and this difference can be further minimized by proper normalization in the training process.

The top row of Table 2 presents the model performance when trained on the color images generated by applying the full pipeline without ISPNet. As we can see, it

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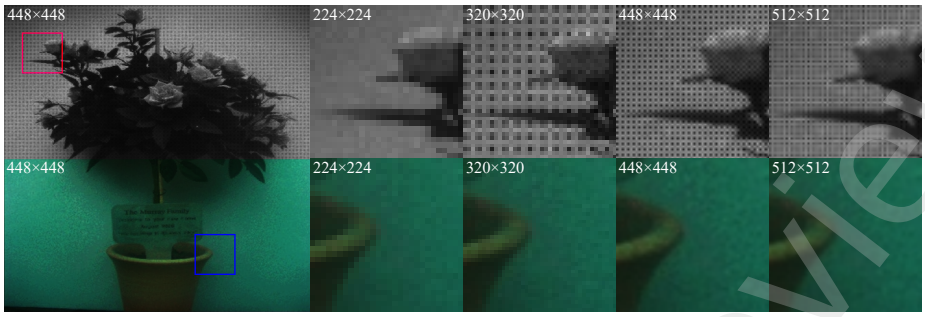

Figure 6: Resize a Bayer pattern image and a demosaiced image to different sizes.

Table 3: Experimental results of classification tasks on ViT-B/16

Training Strategy	Configuration	Top-1 acc.	Top-5 acc.	Data Augmentation
Without ISPNet	Keep all	82.88%	96.88%	✓
With ISPNet	Keep Demosaicing	83.07%	96.841%	✓
	bypass all	83.74%	97.0%	✗

Data augmentation including random resize, clip, flipping, etc.

achieves the lowest top-1 and top-5 accuracy, and all the results in the table show a trend that the more stages applied, the worse the model's performance, indicating that too much ISP processing will undermine the model performance. For the images processed by demosaicing, this degeneration can be compensated to some extent by employing data augmentation.

Based on the above experimental results, we further validated the effectiveness of ISPnet on state-of-the-art image classification models such as ViT-B/16 [36]. All these experiments were trained from scratch for 300 epochs to ensure stable convergence, given the Transformer-based architecture. The corresponding results are summarized in Table 3. It can be observed that the experimental results on ViT-B/16 are generally consistent with those obtained on ResNet-50, indicating the robustness of ISPnet across different backbone architectures.

Configuration Results on Object Detection. In this experiment, two frameworks, Faster-RCNN and Yolo-v3, are used for object detection tasks. The ISPNet is trained from scratch with the detection models together, and the configuration results are pre-

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(a) (b) (c)

Figure 7: Images processed by different ISP pipelines. (a) An original image from ImageNet. The Bayer pattern image processed by (b) the full ISP pipeline in Table 1 and (c) demosaicing and auto white balance only.

Table 4: Configuration results of object detection tasks

Framework	Scheme	Configuration	AP	AP_{50}	AP_{75}
Faster R-CNN	Without ISPNet	Keep All	67.30%	95.60%	79.90%
	With ISPNet	Keep Demosaicing and Histigram Equalization	69.60%	95.20%	83.30%
Yolo-3	Without ISPNet	Keep All	73.13%	95.60%	87.60%
	With ISPNet	Keep Demosaicing	73.20%	96.25%	87.91%

sented in Table 4. It can be found that the overall average precision (AP) on images processed by the ISP pipeline determined by the ISPNet is slightly better than the images processed by the full ISP pipeline, especially for strict metrics (e.g., AP_{75}). Both frameworks preserve demosaicing, with the Faster-RCNN-based framework preserving histogram equalization as well. For demosaicing, as shown in the previous analysis, it is required if there are pre-processing and data augmentation in the training process. The histogram equalization is a conventional enhancement operation for increasing the global contrast of images by adjusting the image intensity distribution.

Fig. 8 presents the images that apply only demosaicing (Fig. 8(a)), demosaicing and histogram equalization (Fig. 8(b)) and the full ISP pipeline (Fig. 8(c)). The images only processed by demosaicing seem 'dark and green' visually, as shown in Fig. 8(a). This can be explained by the histogram on the right. The majority of pixels in this image are distributed in the left part of the histogram, indicating that their values are very small, while the curve of the green channel is slight to the right, indicating that the values of green pixels are larger than others. This is the reason why the image

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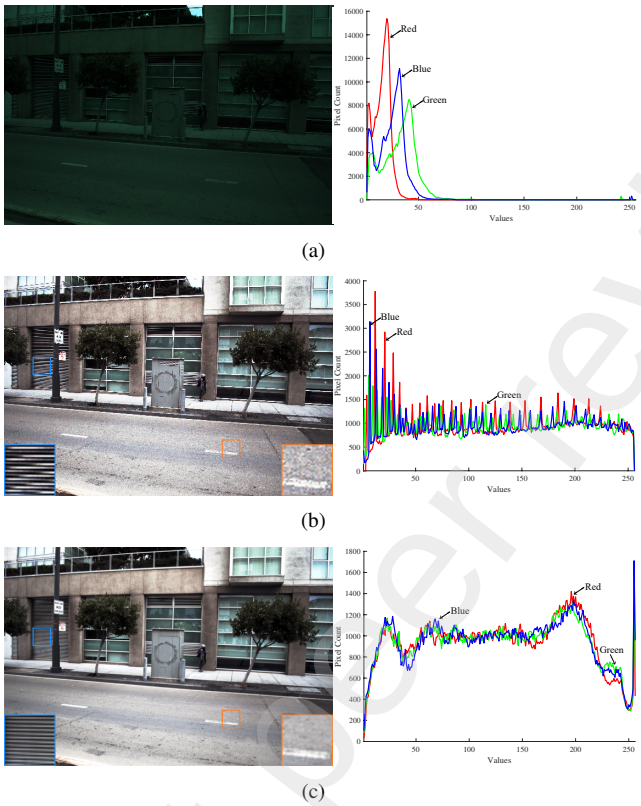

Figure 8: The images apply (a) only demosaicing, (b) demosaicing and histogram equalization and (c) the full ISP with corresponding gray value histogram.

looks 'green'. After applying histogram equalization, the pixels are distributed more equally among $[0, 255]$, which boosts image contrast while playing a similar role to white balance and tone mapping in the ISP pipeline. Compared to Fig. 8(c), there are color noises and artifacts in Fig. 8(b). The images only apply histogram equalization also off-color at times, mainly green. The noise, artifacts and off-color are unfriendly to HVS, but appear to have no impact on computer vision. Apart from the difference in methodology, Yolo-v3 contains more data augmentation technologies than Faster-RCNN, such as multi-scale training and random affine to extend the training dataset, which may explain why the histogram equalization is not necessary for the Yolo-v3 framework.

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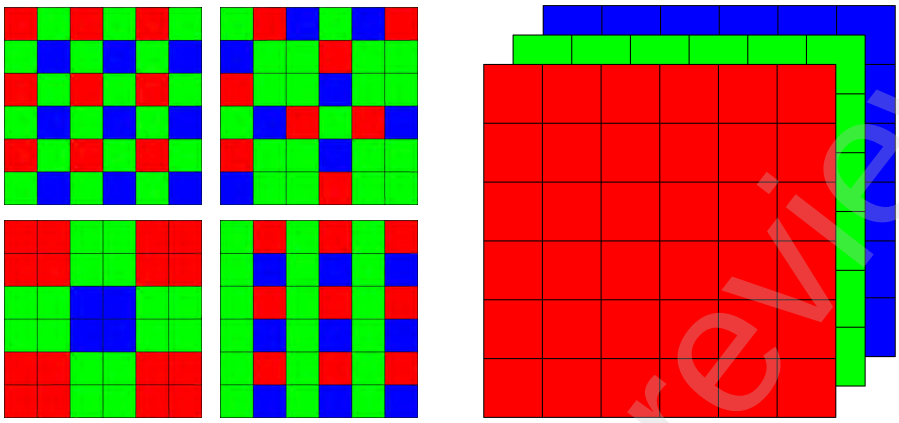

(a) (b)

Figure 9: (a) Different color filter arrays. Top-left: the *RGGB* Bayer pattern, top-right: X-trans pattern, bottom-left: the quad-Bayer pattern and bottom-right: the Yamanaka pattern. (b) The demosaiced three-channel color image.

5.2. Influence of Demosaicing

Demosaicing is a crucial step in converting RAW Bayer pattern images to three-channel images, enabling them to be displayed as color images. Previous experiments have demonstrated that in some cases, the ISP pipeline can be completely bypassed, allowing RAW images to be used directly in computer vision applications. With demosaicing, the model can better resist the artifacts induced by interpolation directly on RAW Bayer pattern images. In practical applications, the situation will be more complicated. For example, the RAW images with high resolution cannot be directly fed to the DNN models for training due to the limited size of GPU memory. Moreover, data augmentation is a key technique to improve the performance of DNN models, especially when the amount of training data is limited. Therefore, in these situations, demosaicing is necessary. The following experiments will further explore the advantages of demosaicing.

5.2.1. Different CFAs

There are various image sensors and each sensor may use a different CFA. For example, there are four types of arrangement in Bayer pattern CFAs, i.e., *RGGB*, *BGGR*, *GRBG* and *GBRG*, Fig. 9(a) presents some examples. To evaluate the generalization

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Table 5: Classification performance when training and evaluating on different Bayer patterns

Training on	Evaluation on	Top-1 acc.	Top-5 acc.
RGGB	*RGGB*	78.21%	94.40%
	BGGR	66.75%	88.94%
	GRBG	43.52%	69.64%
	GBRG	44.29%	70.53%
Demosiaced *RGGB*	Demosiaced *RGGB*	78.17%	94.46%
	Demosiaced *BGGR*	77.57%	94.23%
	Demosiaced *GRBG*	77.63%	94.23%
	Demosiaced *GBRG*	77.48%	94.22%

performance of the model across different CFAs, we train models on *RGGB* pattern images and evaluate them on other Bayer patterns. Note that the *GRBG*, *GBRG* and *BGGR* patterns can be generated from *RGGB* pattern by rearranging a row or/and a column, e.g., the *GRBG* pattern can be generated from the *RGGB* pattern by moving the first column of pixels to the last column.

The first 4 rows in Table 5 present the results of directly training the model on *RGGB* RAW Bayer pattern images and evaluating it on other patterns. It can be found that when the Bayer patterns of training and evaluation dataset are different, the model performance dropped dramatically. Note that these patterns are all Bayer patterns, with just one row and one column different in this experiment. Nevertheless, the images acquired by other CFAs presents in Fig. 9(a) (except for the Bayer CFA) may produce a significant decrease in model performance.

Then these Bayer pattern images are demosaiced with the same algorithm and the same experiments are repeated. The evaluation results are shown in the last four rows of Table 5. It can be found that after demosaicing, there is negligible performance degradation when the training datasets and evaluation datasets are captured by different CFAs. Fig. 9 illustrates this phenomenon. The RAW images captured by different CFAs (Fig. 9(a)) will be converted to three-channel RGB images (Fig. 9(b)) after proper demosaicing. In this case, the demosaicing stage plays a role as 'normalization',

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Table 6: Classification performance when training and evaluating on images processed by different demosaicing methods

Training on	Evaluation on	Top-1 acc.	Top-5 acc.
Hybrid Interpolation	Nearest Neighbor (NN)	67.07%	88.26%
	Linear Interpolation (LI)	71.61%	91.06%
	Cubic Interpolation (CI)	68.06%	89.23%
	Adaptive Color Plane Interpolation (ACPI)	77.57%	94.21%
	Gradient-corrected Linear Interpolation (GLI)	77.76%	94.28%
	Hybrid Interpolation (HI)	78.17%	94.46%

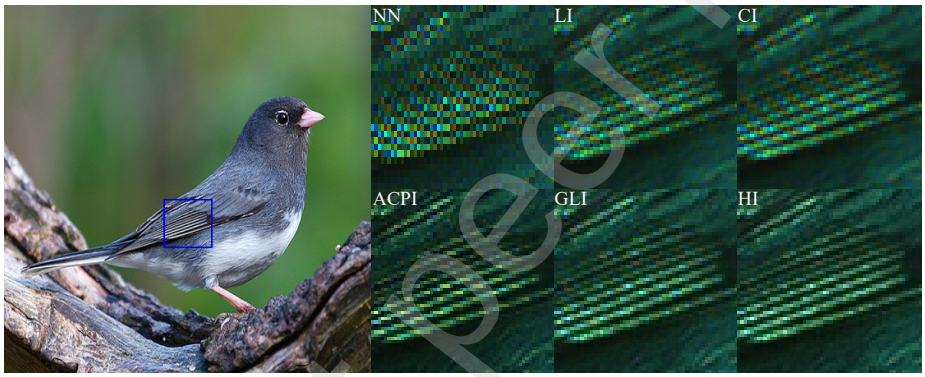

Figure 10: Comparison of different demosaicing methods.

bringing diverse CFA images into similar RGB color images, which leads to better model generalization.

5.2.2. Different Demosaicing Methods

The purpose of this experiment is to figure out whether different demosaicing algorithms have an impact on model performance or not. In our experiments, 6 demosaicing methods of 2 categories are evaluated. The first category covers general interpolation methods including the first three algorithms (NN, LI and CI) in Table 6, which can be used directly for other interpolation tasks. The other includes methods designed specifically for demosaicing (ACPI, GLI and HI), these methods are usually based on the correlation between image channels, such as the color ratio constancy assumption or the color difference constancy assumption. For these dedicated methods, the green

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channel is first interpolated in the demosaicing process, then the interpolated green channel is used as an auxiliary to interpolate R/B channels.

Table 6 presents the results when training the model on the images demosaiced by HI and evaluating it on the datasets demosaiced by the other five algorithms. It can be found that for the first three general interpolation methods, the model performance drops significantly, especially for NN. For the other three interpolation methods specifically designed for demosaicing (ACPI, GLI and HI), the performance degeneration is negligible. Note that different demosaicing methods have an obvious impact on image quality, particularly in texture-rich areas, as shown in Fig 10. The images demosaiced by dedicated methods are generally of higher quality than those demosaiced by generic approaches, with fewer artifacts around the edges, hence the model performance of the images demosaiced by these methods is comparable. Therefore, to assure the model's generalization capability, it is preferable to use interpolation approaches based on the same assumption.

5.3. Results on Degraded Images

The results in Sec. 5.1 show that the policy network converges to one policy, and excludes any outside interference, the majority of ISP's stages can be bypassed. This may be because the images we used in these experiments are taken under good conditions, i.e., the images in the PascalRAW dataset are captured all by Nikon D3200 with favorable lighting.

To figure out whether the policy network may converge to different policies when data vary, e.g., under low-light conditions, we adjust the image luminance by a random gamma transformation. Since images captured under low-light conditions always have heavy noise, we also add Gaussian noise with $\mu = 0, \sigma = 0.05$ to these images, which are presented as (15)

$$I_{\text{deg}}(x, y) = I^{\gamma}(x, y) + G(\mu = 0, \sigma = 0.05). \tag{15}$$

Here, I and I_{deg} are the original and degraded images image pixel values range in $[0, 1]$, $\gamma \in [1, 4]$ is used to compress pixel value, G is the Gaussian function to generate random noise. The detection results of the Yolo-v3 framework under different

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configuration policies are presented in Table 7. It can be found that noise will affect the model performance heavily, and after the full ISP process, the model performance can be improved, while the dynamic configuration of ISP outperforms both of these two configurations.

Fig. 11 presents the application frequency of each stage in the dynamic configuration. As we can see, apart from demosaicing, about 50% and more than 90% images choose to keep the Bayer/color noise reduction stage respectively. These results indicate that noise reduction indeed can improve object detection performance, but not needed for all the input. The influence of the noise is hard to quantify because it not only depends on the strength of the noise, but also affected by the object itself. In addition, although the contrast of images is randomly adjusted by reverse gamma compression ($\gamma > 1$), the policy network does not preserve the gamma compress stage in the ISP pipeline, which verifies that the gamma compression stage is just an enhancement method for HSV. To adjust image brightness under extreme conditions, the adaptive histogram equalization (about 0.1% chosen in Fig. 11) seems more preferred than a fix-parameter gamma compression.

It can also be found that when applying the full ISP process, the detection precision is lower than the dynamic configuration. This result implies that not all operations in the ISP pipeline are beneficial for computer vision tasks. One example is the edge enhancement (sharpen) operation. When the noise in images can be almost completely removed, the sharpening operation can enhance image sharpness for better visual effects, and these effects may not always contribute to computer vision performance (about 5% chosen in Fig. 11). But under extreme conditions, the pixel intensity and noise may be close to each other and the noise is hard to remove. In this situation, applying the edge enhancement processing may further enlarge the impact of noise for computer vision tasks.

5.4. Power Consumption of the ISP Pipeline

Generally, a computer vision system can be divided into two components: the front-end imaging system and the backend vision system. The imaging system includes an image sensor and an image signal processor. It has been shown in [17] that the sensor

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Table 7: The detection performance of the Yolo-v3 framework under different configurations.

Configuration	AP	AP_{50}	AP_{75}
Demosaicing Only	42.87%	69.95%	46.79%
Full ISP	44.93%	72.24%	49.21%
Dynamic	45.93%	74.30%	50.78%

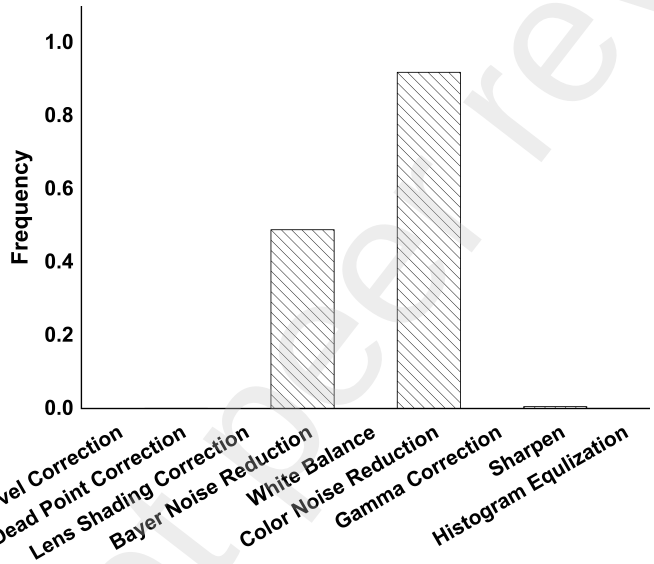

Figure 11: ISP configuration results with degraded images.

part and the ISP part have comparable total power costs. The results of a simulated ISP in [37] show that it consumes 250 mW when processing 1080p video at 60 fps, which is comparable to backend computer vision processors. Therefore, if we can skip the ISP pipeline, the computational complexity and power consumption of the entire computer vision system can be reduced significantly. Moreover, [38] proposes an end-to-end HOG+SVM accelerator that takes RAW images as input. This design can achieve comparable detection performance with color image-based accelerators while saving a significant amount of power (2.5 $\times$ of the accelerator). With our analysis above, a similar method can be further extended to deep learning-based accelerators.

Our experiments also show that under some extreme conditions (i.e., noisy heavy

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Table 8: Resource consumption of each ISP module in [39].

Module	LUT	DFF	Slice	Memory
Dead Point Correction	759	609	330	4 line buff
Joint Demosaicing, Denoise and Sharpen	8182	4115	2709	4 line buff
White Balance	2243	1999	956	3×18kb
Gamma Correction	1964	0	646	0
Color Correction	826	470	271	0

situations), some stages in the ISP pipeline will improve the model performance in object detection while most of the stages can still be bypassed. [39] proposed a well-designed ISP pipeline and presents the resource consumption of each module on FPGA implementation, some of them are shown in Table 8. It can be found that even under extreme conditions, bypassing some unnecessary steps can still result in resource savings.

6. Discussion

The conventional design of HVS-oriented ISP pipeline is always isolated from the computer vision tasks, which is unfavorable to dedicated vision system design. The imaging system, including the ISP pipeline, has an impact on not only vision model performance, but also on the power consumption, efficiency, cost, etc., of the entire vision system. In this paper, a hybrid supervised learning and reinforcement learning-based framework is proposed to study the optimal configuration of computer vision-oriented ISP pipelines. Moreover, this framework can be further extended to study the influence of other classical image processing methods on learning-based tasks, bridging the gap between classical methods and deep learning-based methods.

The ISP pipeline, to our understanding, serves as a 'normalization' or 'standardization' mechanism for both HVS and computer vision tasks. For HVS, ISP pipelines designed for producing pleasurable images, although aesthetics vary, several common fundamental criteria are shared such as fewer artifacts, fewer noises, accurate color,

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etc. Therefore, the 'variance' of the data gets smaller and smaller during the processing in order to make it easier for the HVS to accept. This is similar in machine learning, where data with smaller variances are easier to be fitted and obtain statistical patterns. However, not all the stages in the ISP pipeline are necessary for computer vision tasks, because there is no obvious connection between the requirements of HVS and computer vision. Furthermore, too much processing may potentially impair the original information of the captured RAW images.

Our experiments yield some interesting findings. Ideally, the RAW Bayer pattern images would suffice to conduct follow-up computer vision tasks, which means the ISP pipeline can be totally bypassed to some extent. The model trained on Bayer pattern images, on the other hand, has poor generalization performance, inability to generalize across different CFAs, and to resist the artifacts formed by image-size-changing interpolation. This is limited by the special structure of CFAs, which store the image's color information in a minimalistic manner, but result in pixel discontinuities. Demosaicing is a straightforward method for correcting these concerns since it recovers the color information of images while keeping the pixels continuous among each color channel. Therefore, demosaicing may be regarded as a 'normalizing' method that brings together various patterns into one space and decreases data variance, which is beneficial for feature extraction. When preprocessing is required for a task, performing demosaicing beforehand can help to prevent many of the issues that arise when preprocessing directly on Bayer pattern images. As for the dynamic configuration, its advantages can mainly be demonstrated under extreme conditions, in most scenes under normal shooting conditions, simply bypassing the ISP or applying a demosaicing seems sufficient for computer vision tasks.

7. Conclusion

To find the optimal ISP configuration for computer vision tasks, in this paper, an associative search-based policy network is proposed to obtain the ISP configuration policy for a given input RAW Bayer image. This network is optimized together with the corresponding computer vision tasks, aiming at improving the overall vision performance. Experimental results show that the model trained on the Bayer pattern images

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achieves the best top-1 performance among different configurations on the ResNet50 classification challenge, meaning that the Bayer pattern images contain all the information needed by computer vision tasks and the ISP pipeline is redundant for this task. In addition, the experiments on object detection illustrate that the demosaicing step is necessary for tasks with dedicated preprocessing and/or data augmentation, while visual enhancement algorithms such as image histogram equalization appear to help as well in the cases with small-scale training datasets. Demosaicing can not only preserve images' color information but also reduce artifact generation during processing. In addition, it can also improve the generalization ability of models across various CFAs. In general, for computer vision tasks, the ISP pipeline, or most of the ISP stages, can be eliminated to reduce the computational complexity and power consumption of the entire vision system, without sacrificing the vision performance.

References

[1] Ramanath, R., Snyder, W.E., Yoo, Y., Drew, M.S., 2005. Color image processing pipeline. IEEE Signal Processing Magazine 22, 34–43.

[2] Li, X., Liu, M., Ling, Q., 2024. Pixel-wise gamma correction mapping for low-light image enhancement. IEEE Transactions on Circuits and Systems for Video Technology 34, 681–694. doi:10.1109/TCSVT.2023.3286802.

[3] Zwislocki, J.J., 2009. Stevens' power law. Sensory Neuroscience: Four Laws of Psychophysics , 1–80.

[4] Privitera, A.J., 2018. Sensation and perception. General Psychology FA18 , 64.

[5] Chakrabarti, A., Scharstein, D., Zickler, T.E., 2009. An empirical camera model for internet color vision, in: Proc. British Machine Vision Conference, p. 4.

[6] Piciarelli, C., Esterle, L., Khan, A., Rinner, B., Foresti, G.L., 2016. Dynamic reconfiguration in camera networks: A short survey. IEEE Transactions on Circuits and Systems for Video Technology 26, 965–977. doi:10.1109/TCSVT.2015.2426575.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5367628

[7] Jang, E., Gu, S., Poole, B., 2017. Categorical reparameterization with gumbel-softmax, in: Proc. International Conference on Learning Representations.

[8] Srivastava, R.K., Greff, K., Schmidhuber, J., 2015. Highway networks, in: Proc. International Conference on Machine Learning workshop on deep learning.

[9] Wang, X., Yu, F., Dou, Z.Y., Darrell, T., Gonzalez, J.E., 2018. Skipnet: Learning dynamic routing in convolutional networks, in: Proc. European Conference on Computer Vision, pp. 409–424.

[10] Wu, Z., Nagarajan, T., Kumar, A., Rennie, S., Davis, L.S., Grauman, K., Feris, R., 2018. Blockdrop: Dynamic inference paths in residual networks, in: Proc. IEEE Conference on Computer Vision and Pattern Recognition, pp. 8817–8826.

[11] Zini, S., Buzzelli, M., 2025. Bayesian nights: Optimizing night photography rendering with bayesian derivative-free methods. Pattern Recognition 161, 111314. URL: https://www.sciencedirect.com/science/article/pii/S0031320324010653, doi:https://doi.org/10.1016/j.patcog.2024.111314.

[12] Bai, C., Liu, F., Li, J., 2025. Joint learning of rgbw color filter arrays and demosaicking. Pattern Recognition 157, 110929. URL: https://www.sciencedirect.com/science/article/pii/S0031320324006800, doi:https://doi.org/10.1016/j.patcog.2024.110929.

[13] Mosleh, A., Sharma, A., Onzon, E., Mannan, F., Robidoux, N., Heide, F., 2020. Hardware-in-the-loop end-to-end optimization of camera image processing pipelines, in: Proc. IEEE Conference on Computer Vision and Pattern Recognition.

[14] Zhou, W., Zhang, L., Gao, S., Lou, X., 2021. Gradient-based feature extraction from raw bayer pattern images. IEEE Transactions on Image Processing 30, 5122–5137. doi:10.1109/TIP.2021.3067166.

[15] Dai, L., Liu, H., Song, P., Liu, M., 2024. A gated cross-domain collaborative network for underwater object detection. Pattern Recognition 149,

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5367628

110222. URL: https://www.sciencedirect.com/science/article/pii/S0031320323009196, doi:https://doi.org/10.1016/j.patcog.2023.110222.

[16] Gil Rodríguez, R., Vazquez-Corral, J., Bertalmío, M., Finlayson, G.D., 2024. Color matching in the wild. Pattern Recognition 154, 110575. URL: https://www.sciencedirect.com/science/article/pii/S0031320324003261, doi:https://doi.org/10.1016/j.patcog.2024.110575.

[17] Buckler, M., Jayasuriya, S., Sampson, A., 2017. Reconfiguring the imaging pipeline for computer vision, in: Proc. IEEE International Conference on Computer Vision, pp. 975–984.

[18] Li, Z., Lu, M., Zhang, X., Feng, X., Asif, M.S., Ma, Z., 2024. Efficient visual computing with camera raw snapshots. IEEE Transactions on Pattern Analysis and Machine Intelligence , 46.

[19] Yu, K., Li, Z., Peng, Y., Loy, C.C., Gu, J., 2021. Reconfigisp: Reconfigurable camera image processing pipeline, in: Proc. International Conference on Computer Vision, pp. 4248–4257.

[20] Chen, C., Qifeng, C., Jia, X., Vladlen, K., 2018. Learning to see in the dark, in: Proc. the IEEE International Conference on Computer Vision.

[21] Abdelrahman, A., Stephen, L., Michael, S.B., . Smartphone image denoising dataset. [Online]. Available:https://www.eecs.yorku.ca/~kamel/sidd/. Accessed January 25, 2022.

[22] Franzen, R.W., . True color kodak images. [Online]. Available:http://r0k.us/graphics/kodak/. Accessed September 10, 2019.

[23] Laurent, C., . Laurent condat's image base. [Online]. Available:https://lcondat.github.io/imagebase.html. Accessed January 25, 2022.

[24] Zhang, L., Wu, X., Antoni, B., Li, X., . Mcmaster dataset. [Online]. Available:http://www4.comp.polyu.edu.hk/~cslzhang/CDM_Dataset.htm. Accessed January 25, 2022.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5367628

[25] Omid-Zohoor, A., Young, C., Ta, D., Murmann, B., 2018. Toward always-on mobile object detection: Energy versus performance tradeoffs for embedded hog feature extraction. IEEE Transactions on Circuits and Systems for Video Technology 28, 1102–1115.

[26] Lin, H., Kim, S.J., Süsstrunk, S., Brown, M.S., 2011. Revisiting radiometric calibration for color computer vision, in: 2011 International Conference on Computer Vision, pp. 129–136.

[27] Zhou, W., Zhang, X., Wang, H., Gao, S., Lou, X., 2024. Raw bayer pattern image synthesis for computer vision-oriented image signal processing pipeline design, in: arXiv.

[28] Han, Y., Huang, G., Song, S., Yang, L., Wang, H., Wang, Y., 2021. Dynamic neural networks: A survey. IEEE Transactions on Pattern Analysis and Machine Intelligence .

[29] Cui, S., Li, D., Zhang, J., 2025. Dynamic multi-scale feature augmentation for inductive network representation learning. Pattern Recognition 161, 111250. URL: https://www.sciencedirect.com/science/article/pii/S003132032401001X, doi:https://doi.org/10.1016/j.patcog.2024.111250.

[30] Omid-Zohoor, A., Ta, D., Murmann, B., . PASCALRAW: Raw image database for object detection. [Online]. Available:http://purl.stanford.edu/hq050zr7488. Accessed September 11, 2019.

[31] Duc-Tien, D.N., Cecilia, P., Valentina, C., Giulia, B., . Raise RAW image dataset. [Online]. Available:http://loki.disi.unitn.it/RAISE/. Accessed January 25, 2022.

[32] Ignatov, A., Timofte, R., Ko, S.J., et al., 2019. AIM 2019 challenge on raw to rgb mapping: Methods and results, in: Proc. IEEE/CVF International Conference on Computer Vision Workshop, pp. 3584–3590. doi:10.1109/ICCVW.2019.00443.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5367628

[33] Sutton, R.S., Barto, A.G., 2018. Reinforcement learning: An introduction. MIT press.

[34] Williams, R.J., 1992. Simple statistical gradient-following algorithms for connectionist reinforcement learning. Machine learning 8, 229–256.

[35] Mnih, V., Badia, A.P., Mirza, M., Graves, A., Lillicrap, T., Harley, T., Silver, D., Kavukcuoglu, K., 2016. Asynchronous methods for deep reinforcement learning, in: Proc. of The International Conference on Machine Learning, pp. 1928–1937.

[36] Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., et al., 2020. An image is worth 16x16 words: Transformers for image recognition at scale. arXiv preprint arXiv:2010.11929 .

[37] Hegarty, J., Brunhaver, J., Devito, Z., Ragankelley, J., Cohen, N., Bell, S., Vasilyev, A., Horowitz, M., Hanrahan, P., 2014. Darkroom:compiling high-level image processing code into hardware pipelines. Acm Transactions on Graphics 33, 1–11.

[38] Zhang, X., Zhang, L., Lou, X., 2022. A raw image-based end-to-end object detection accelerator using hog features. IEEE Transactions on Circuits and Systems I: Regular Papers 69, 322–333. doi:10.1109/TCSI.2021.3098053.

[39] Jin, W., He, G., He, W., Mao, Z., 2017. A 12-bit 4928 × 3264 pixel cmos image signal processor for digital still cameras. Integration 59, 206–217.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5367628